Milestone 1 Report

PCAP Conversion and Network Traffic Data Preparation

Project Title

Hands-On PCAP Analysis and Threat Detection Using Generative AI

1. Objective

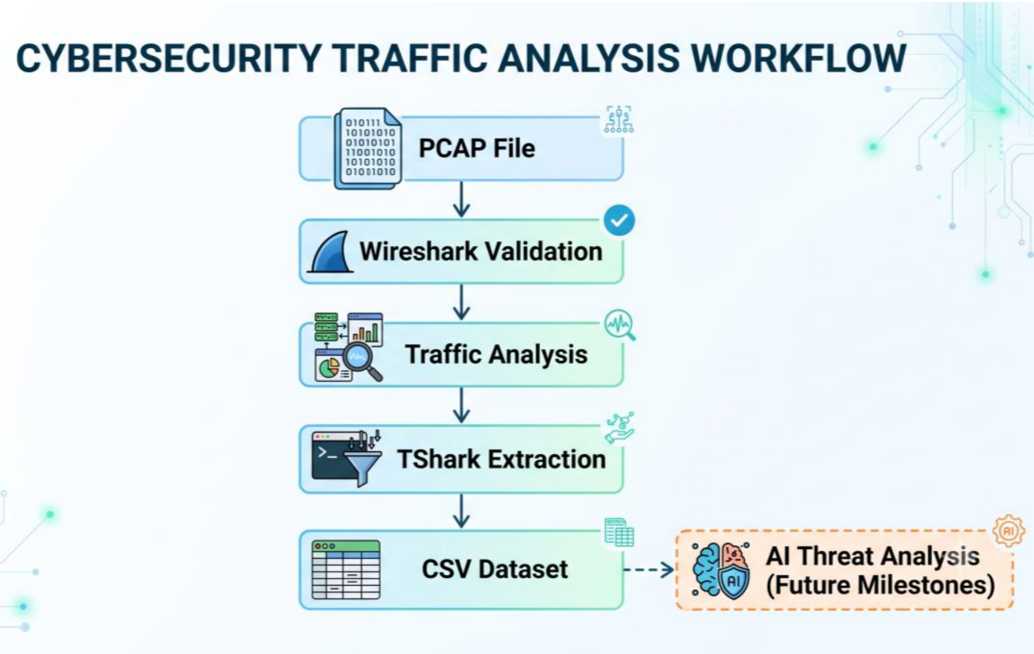
The objective of this milestone was to prepare the provided PCAP (Packet Capture) file for cybersecurity analysis by validating the captured network traffic, extracting important packet information, and converting the raw packet data into a structured CSV dataset.

The main objectives of this milestone were:

- Set up the required packet analysis environment.
- Validate the provided PCAP file using Wireshark.
- Analyze initial traffic statistics and protocol distribution.
- Extract network traffic information using TShark.
- Convert packet-level data into a structured CSV format for further AI-assisted threat analysis.

The prepared dataset will be used in later milestones for identifying suspicious communication patterns, analyzing network behavior, and supporting threat detection activities.

The overall workflow followed during this milestone is illustrated below:



2. Tools and Technologies Used

The following tools were used during this milestone:

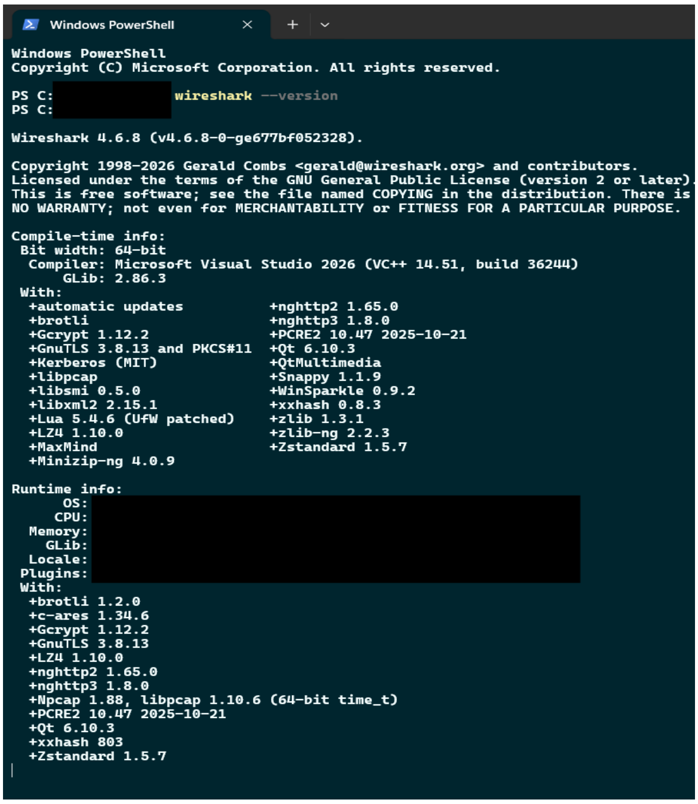
Tool	Purpose
Wireshark	Used for graphical packet inspection, PCAP validation, and protocol analysis
TShark	Used for command-line extraction of packet information from PCAP files
Microsoft PowerShell	Used for executing TShark commands and managing analysis files
Microsoft Excel	Used for validating and reviewing the generated CSV dataset
Windows 11	Operating system used for performing the analysis

Evidence: Tool Installation and Verification

Wireshark and TShark were installed and verified before beginning PCAP analysis.

The installed versions were confirmed using PowerShell commands:

```
wireshark --version
tshark --version
```



```
Windows PowerShell
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PS C:[redacted] wireshark --version
PS C:[redacted]

Wireshark 4.6.8 (v4.6.8-0-ge677bf052328).

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NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
 Bit width: 64-bit
  Compiler: Microsoft Visual Studio 2026 (VC++ 14.51, build 36244)
      GLib: 2.86.3
 With:
  +automatic updates            +nghttp2 1.65.0
  +brotli                       +nghttp3 1.8.0
  +Gcrypt 1.12.2                +PCRE2 10.47 2025-10-21
  +GnuTLS 3.8.13 and PKCS#11    +Qt 6.10.3
  +Kerberos (MIT)               +QtMultimedia
  +libpcap                      +Snappy 1.1.9
  +libsmi 0.5.0                 +WinSparkle 0.9.2
  +libxml2 2.15.1               +xxhash 0.8.3
  +Lua 5.4.6 (UfW patched)      +zlib 1.3.1
  +LZ4 1.10.0                   +zlib-ng 2.2.3
  +MaxMind                      +Zstandard 1.5.7
  +Minizip-ng 4.0.9

Runtime info:
      OS:
     CPU:
  Memory:
    GLib:
  Locale:
 Plugins: [redacted]
 With:
  +brotli 1.2.0
  +c-ares 1.34.6
  +Gcrypt 1.12.2
  +GnuTLS 3.8.13
  +LZ4 1.10.0
  +nghttp2 1.65.0
  +nghttp3 1.8.0
  +Npcap 1.88, libpcap 1.10.6 (64-bit time_t)
  +PCRE2 10.47 2025-10-21
  +Qt 6.10.3
  +xxhash 803
  +Zstandard 1.5.7
```
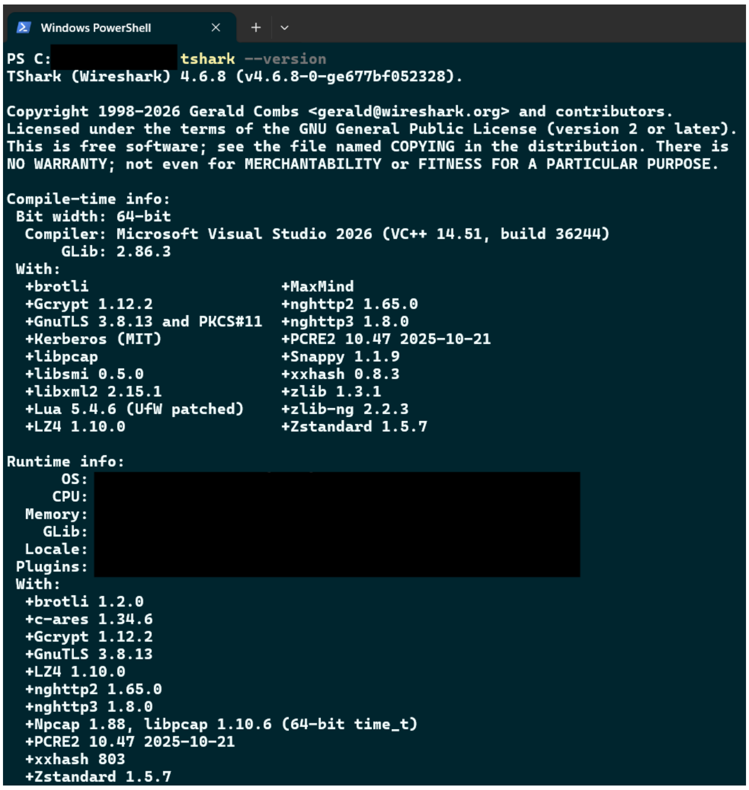

Figure 1: Wireshark installation verification showing the installed Wireshark version.



```
PS C:[redacted] tshark --version
TShark (Wireshark) 4.6.8 (v4.6.8-0-ge677bf052328).

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  Compiler: Microsoft Visual Studio 2026 (VC++ 14.51, build 36244)
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  +GnuTLS 3.8.13 and PKCS#11    +nghttp3 1.8.0
  +Kerberos (MIT)               +PCRE2 10.47 2025-10-21
  +libpcap                      +Snappy 1.1.9
  +libsmi 0.5.0                 +xxhash 0.8.3
  +libxml2 2.15.1               +zlib 1.3.1
  +Lua 5.4.6 (UfW patched)      +zlib-ng 2.2.3
  +LZ4 1.10.0                   +Zstandard 1.5.7

Runtime info:
      OS:
     CPU:
  Memory:
    GLib:
  Locale:
 Plugins: [redacted]
 With:
  +brotli 1.2.0
  +c-ares 1.34.6
  +Gcrypt 1.12.2
  +GnuTLS 3.8.13
  +LZ4 1.10.0
  +nghttp2 1.65.0
  +nghttp3 1.8.0
  +Npcap 1.88, libpcap 1.10.6 (64-bit time_t)
  +PCRE2 10.47 2025-10-21
  +xxhash 803
  +Zstandard 1.5.7
```

Figure 2: TShark installation verification confirming command-line packet analysis functionality.

3. PCAP File Information

The provided PCAP file was reviewed before performing any analysis.

The file information is shown below:

Parameter	Details
File Name	2021-08-16-formbook.pcap
File Type	Wireshark Capture File (.pcap)
Format	pcap
Encapsulation	Ethernet
File Size	2251 KB
Total Packets	3588
Capture Duration	610.400 seconds
Capture Start Time	2021-08-17 01:58:06
Capture End Time	2021-08-17 02:08:16

The PCAP file was successfully identified as a valid Wireshark packet capture file.

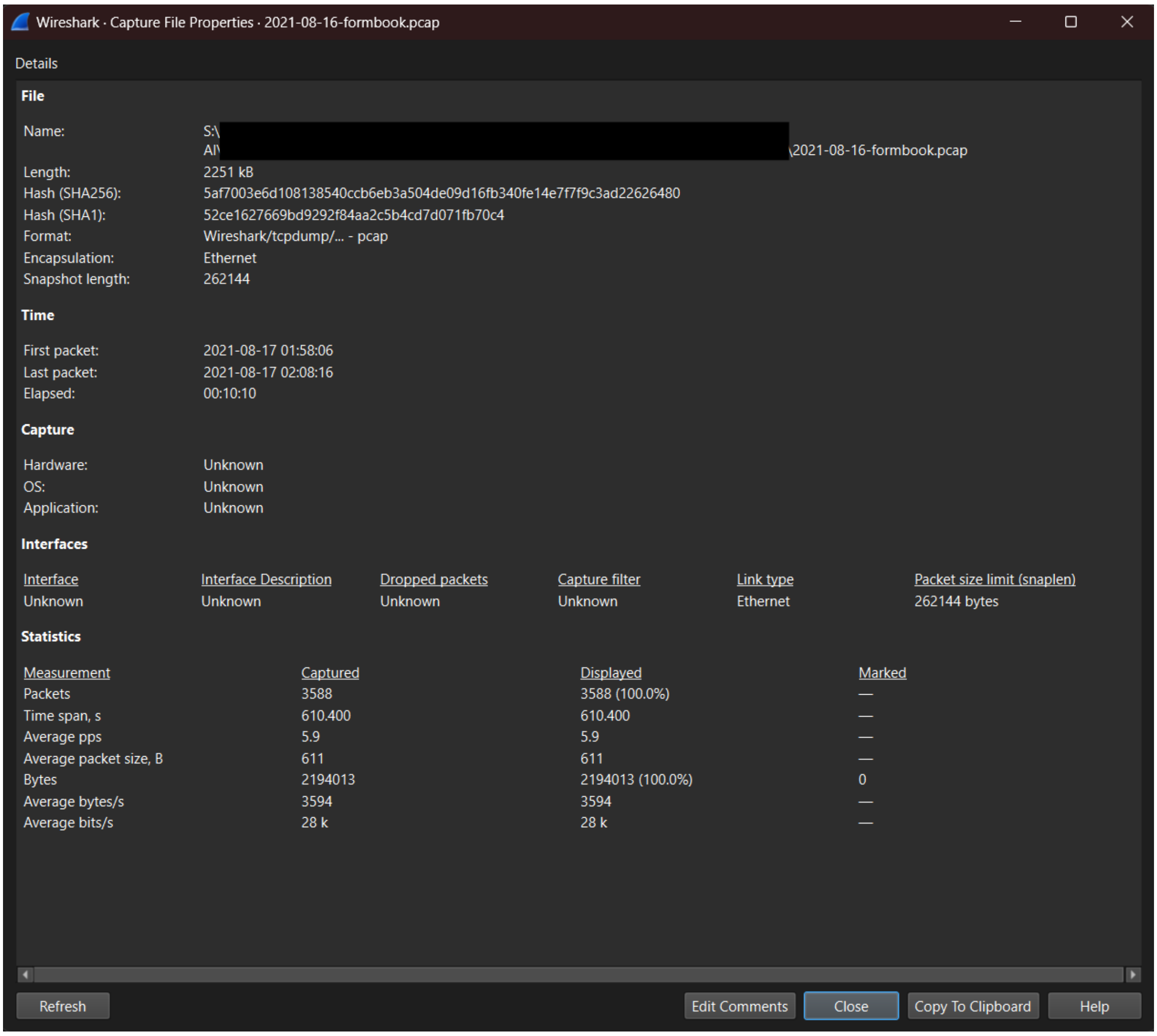


Figure 3: Wireshark Capture File Properties displaying PCAP metadata, packet count, and capture duration.

4. PCAP Validation Using Wireshark

Before extracting data, the PCAP file was opened in Wireshark to confirm that the capture was readable and contained valid network packets.

Wireshark was used because it provides detailed visibility into:

- Individual packets
- Network protocols
- Source and destination information
- Packet timestamps
- Communication patterns

The PCAP opened successfully without any corruption or loading issues.

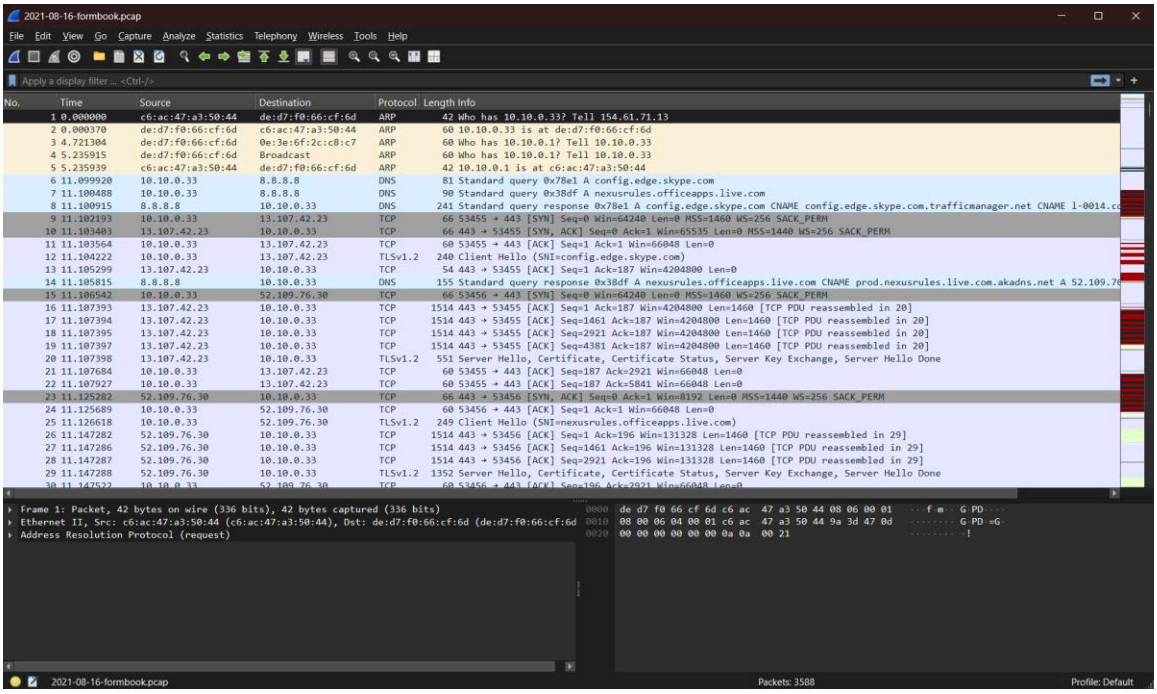


Figure 4: Provided PCAP file successfully opened in Wireshark for initial validation.

5. Initial Traffic Statistics and Protocol Analysis

5.1 Capture Statistics

Initial statistics were collected using Wireshark to understand the size and duration of the captured traffic.

The analysis showed:

Metric	Value
Total Packets	3588
Capture Duration	610.400 seconds
Average Packets Per Second	5.9 packets/sec
Average Packet Size	611 bytes

These statistics provide an overview of the network activity captured inside the PCAP file.

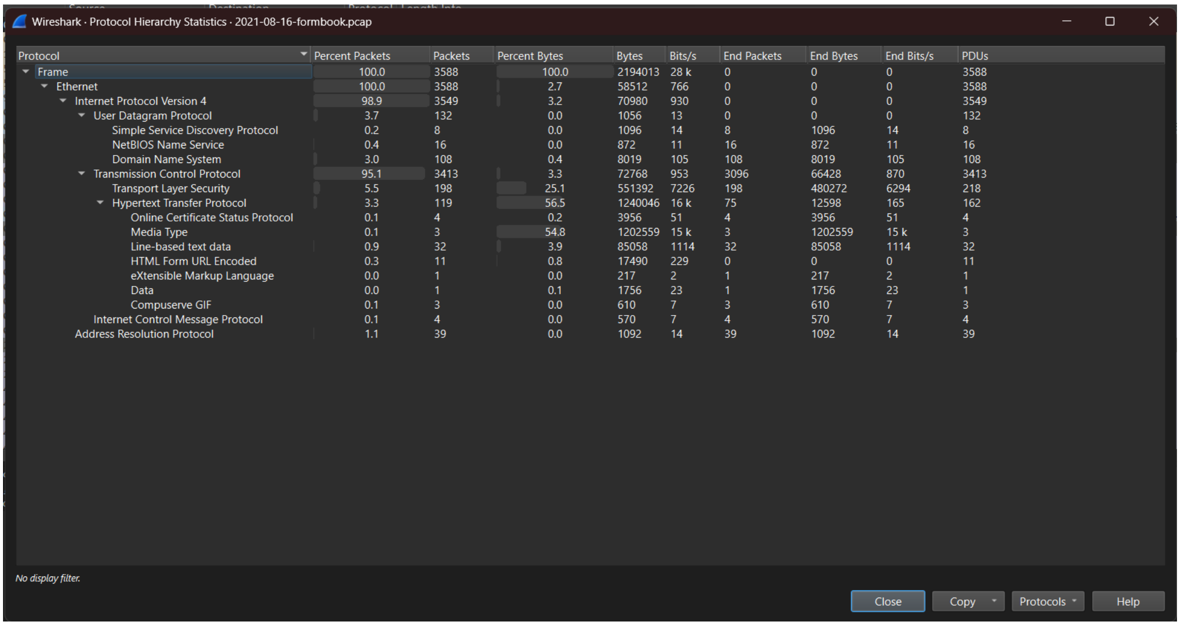
5.2 Protocol Hierarchy Analysis

The Wireshark Protocol Hierarchy feature was used to identify the different protocols present in the captured traffic.

The following protocols were observed:

Protocol	Observation
TCP	Majority of network traffic
UDP	Used for UDP-based communication
DNS	Domain resolution traffic observed
TLS	Encrypted communication observed
HTTP	Web communication observed
ARP	Local network discovery traffic observed

The protocol distribution indicates that the capture contains a mixture of normal network communication protocols that require further investigation during later threat analysis stages.


Wireshark · Protocol Hierarchy Statistics · 2021-08-16-formbook.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	3588	100.0	2194013	28 k	0	0	0	3588
Ethernet	100.0	3588	2.7	58512	766	0	0	0	3588
Internet Protocol Version 4	98.9	3549	3.2	70980	930	0	0	0	3549
User Datagram Protocol	3.7	132	0.0	1056	13	0	0	0	132
Simple Service Discovery Protocol	0.2	8	0.0	1096	14	8	1096	14	8
NetBIOS Name Service	0.4	16	0.0	872	11	16	872	11	16
Domain Name System	3.0	108	0.4	8019	105	108	8019	105	108
Transmission Control Protocol	95.1	3413	3.3	72768	953	3096	66428	870	3413
Transport Layer Security	5.5	198	25.1	551392	7226	198	480272	6294	218
Hypertext Transfer Protocol	3.3	119	56.5	1240046	16 k	75	12598	165	162
Online Certificate Status Protocol	0.1	4	0.2	3956	51	4	3956	51	4
Media Type	0.1	3	54.8	1202559	15 k	3	1202559	15 k	3
Line-based text data	0.9	32	3.9	85058	1114	32	85058	1114	32
HTML Form URL Encoded	0.3	11	0.8	17490	229	0	0	0	11
eXtensible Markup Language	0.0	1	0.0	217	2	1	217	2	1
Data	0.0	1	0.1	1756	23	1	1756	23	1
Compuserve GIF	0.1	3	0.0	610	7	3	610	7	3
Internet Control Message Protocol	0.1	4	0.0	570	7	4	570	7	4
Address Resolution Protocol	1.1	39	0.0	1092	14	39	1092	14	39

Figure 5: Wireshark Protocol Hierarchy showing protocol distribution within the PCAP capture.

6. PCAP Conversion Using TShark

6.1 Conversion Process

After validating the PCAP file, TShark was used to extract packet information and convert the raw packet capture into a structured CSV dataset.

TShark was selected because it enables automated extraction of packet fields from large captures and creates a format suitable for further analysis.

The following fields were extracted:

Field	Purpose
frame.time	Packet timestamp
ip.src	Source IP address
ip.dst	Destination IP address
tcp.srcport	TCP source port
tcp.dstport	TCP destination port
udp.srcport	UDP source port
udp.dstport	UDP destination port
frame.protocols	Protocol information
frame.len	Packet size

6.2 TShark Command Used

The following command was executed:

```
tshark -r "2021-08-16-formbook.pcap" -T fields -e frame.time -e ip.src -e ip.dst -
e tcp.srcport -e tcp.dstport -e udp.srcport -e udp.dstport -e frame.protocols -e
frame.len -E header=y -E separator="," > ..\Converted_Data\formbook_traffic.csv
```

The command performed the following actions:

- Read the PCAP file.
- Extracted selected packet fields.
- Added CSV column headers.
- Generated a structured CSV output file.

Generated output:

formbook_traffic.csv

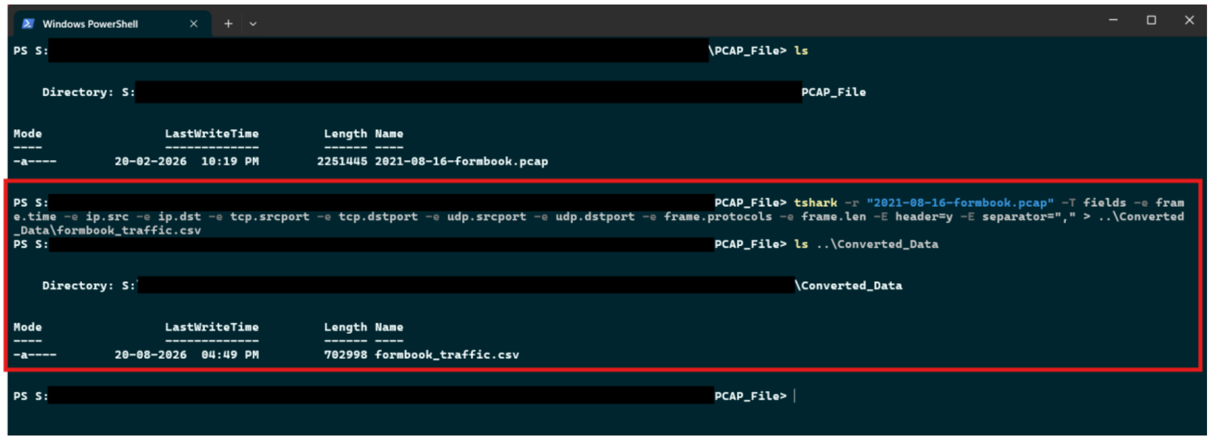


Figure 6: Successful PCAP to CSV conversion using TShark.

7. CSV Data Validation

After conversion, the generated CSV file was validated using Microsoft Excel.

The validation process confirmed:

- CSV headers were created correctly.
- Packet information was separated into individual columns.
- Extracted fields were readable.
- Dataset size matched the original packet count.

The CSV dataset contained:

3588 rows

This matched the original Wireshark packet count:

3588 packets

This confirmed that the packet extraction process completed successfully.

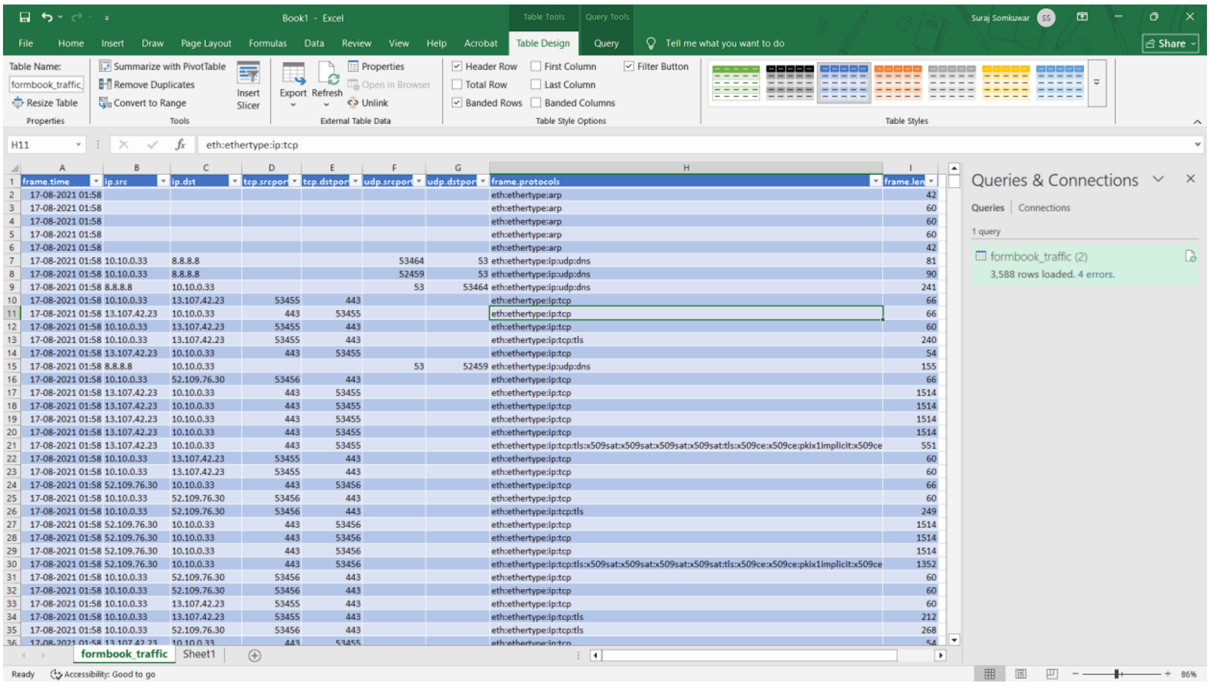


Figure 7: CSV output validation showing structured packet information loaded into Microsoft Excel.

8. Generated Output Files

The following files were generated during this milestone:

File Name	**Purpose**
2021-08-16-formbook.pcap	Original network packet capture file
formbook_traffic.csv	CSV dataset generated using TShark
formbook_traffic_validation.xlsx	Excel validation copy for reviewing extracted traffic data

9. Security Relevance

Preparing structured network traffic data is an important step in cybersecurity investigations.

The extracted dataset can support:

- Identification of suspicious IP communication.
- Analysis of network protocols.
- Investigation of unusual ports.
- Detection of abnormal communication patterns.
- AI-assisted threat analysis in later milestones.

The conversion process transforms raw packet evidence into a dataset that can be analyzed using automated and AI-based techniques.

10. Challenges and Resolution

Challenge: CSV Formatting Issue

During initial CSV review, the generated CSV file opened incorrectly in Excel, with all data appearing in a single column.

Resolution:

The CSV was re-imported using Excel's ***Data → From Text/CSV*** option and the delimiter was correctly selected as a comma.

After correction, the data was displayed in separate columns and successfully validated.

11. Milestone 1 Conclusion

Milestone 1 was successfully completed.

The provided PCAP file was validated using Wireshark, analyzed for initial traffic characteristics, converted into a structured CSV dataset using TShark, and verified using Microsoft Excel.

The prepared dataset contains important network information including:

- timestamps
- source and destination IP addresses
- ports
- protocols
- packet lengths

This prepared dataset will be used in subsequent milestones for AI-assisted threat detection and deeper cybersecurity analysis.